

Pythagoras' theorem

Pythagoras sats

MATH

Discovered 1800 BC · Babylon

SPARKON

RARE

01 / 10

Percent change

Förändringsfaktor

MATH

History 1st century AD · Roman taxes

SPARKON

COMMON

02 / 10

Newton's second law

Newtons andra lag

PHYSICS

Discovered 1687 · Isaac Newton

SPARKON

LEGENDARY

03 / 10

Energy conservation

Energiprincipen

PHYSICS

Discovered 1847 · Mayer and Joule

SPARKON

RARE

04 / 10

Ohm's law

Ohms lag

PHYSICS

Discovered 1827 · Georg Ohm

SPARKON

COMMON

05 / 10

Density

Densitet

CHEMISTRY

Milestone 3rd century BC · Archimedes

SPARKON

COMMON

06 / 10

Oxygen

Syre

CHEMISTRY

Discovered 1772 · C. W. Scheele

SPARKON

COMMON

07 / 10

Gold

Guld

CHEMISTRY

Known Prehistory · oldest jewellery: Varna

SPARKON

LEGENDARY

08 / 10

DNA

DNA

BIOLOGY

Double helix 1953 · Franklin, Watson, Crick

SPARKON

LEGENDARY

09 / 10

Newton's second law



THE IDEA

A total force changes how something moves, and the heavier it is, the less it changes.

$$F = m \cdot a$$

LETS YOU PREDICT

Knowing force and mass, where a thing will be later.

TESTS USUALLY ASK

- Find the acceleration
- Braking or friction force



Scan for the full explanation and a practice question.
SPARKON · 03 / 10

Percent change



THE IDEA

A change is measured against where you started, so you multiply by a change factor.

$$\text{new} = \text{old} \times \text{factor}$$

LETS YOU PREDICT

Where a price ends after n equal changes: $\text{old} \times \text{factor}^n$.

TESTS USUALLY ASK

- Up 20 %, then down 20 %: back to start?
- Interest over several years



Scan for the full explanation and a practice question.
SPARKON · 02 / 10

Pythagoras' theorem



THE IDEA

In a right-angled triangle, the squares on the two short sides add up to the square on the long side.

$$a^2 + b^2 = c^2$$

LETS YOU PREDICT

A distance you cannot measure, from two you can.

TESTS USUALLY ASK

- Find the missing side
- Is this triangle right-angled?



Scan for the full explanation and a practice question.
SPARKON · 01 / 10

Density



THE IDEA

Density is how much mass is packed into each bit of volume. An object floats when its average density is lower than the liquid's.

$$\rho = m / V$$

LETS YOU PREDICT

Whether something floats before you drop it in.

TESTS USUALLY ASK

- Calculate a density
- Which substance is it?



Scan for the full explanation and a practice question.
SPARKON · 06 / 10

Ohm's law



THE IDEA

For an ordinary resistor at constant temperature, the current grows in step with the voltage across it.

$$U = R \cdot I$$

LETS YOU PREDICT

The current before you switch on.

TESTS USUALLY ASK

- Find current, voltage or resistance
- Resistors in series



Scan for the full explanation and a practice question.
SPARKON · 05 / 10

Energy conservation



THE IDEA

Energy never appears or disappears; it only changes form, so the total stays the same.

$$m \cdot g \cdot h = \frac{1}{2} \cdot m \cdot v^2 \text{ (ideal drop)}$$

LETS YOU PREDICT

Speed at the bottom of a drop from rest, if friction is small.

TESTS USUALLY ASK

- Speed of a falling object
- Where did the energy go?



Scan for the full explanation and a practice question.
SPARKON · 04 / 10

DNA



THE IDEA

DNA is a code of four letters on a twisted ladder, and each half tells you the other.

$$\text{A pairs with T} \cdot \text{C pairs with G}$$

LETS YOU PREDICT

One strand gives you the other strand.

TESTS USUALLY ASK

- Write the partner strand
- How a cell copies DNA



Scan for the full explanation and a practice question.
SPARKON · 09 / 10

Gold



THE IDEA

Gold is a noble metal: it hardly reacts, so it never rusts and stays shiny for thousands of years.

$$\text{Au} \cdot \text{atomic number } 79 \cdot \rho = 19.3 \text{ g/cm}^3$$

LETS YOU PREDICT

Treasure found after 3,000 years still shines.

TESTS USUALLY ASK

- Reactivity: why gold does not rust
- Carats as percent



Scan for the full explanation and a practice question.
SPARKON · 08 / 10

Oxygen



THE IDEA

Oxygen has 6 outer electrons and wants 2 more, so it reacts with almost everything.

$$\text{O} \cdot \text{atomic number } 8 \cdot \text{shells } 2, 6$$

LETS YOU PREDICT

It gains 2 electrons (O^{2-}) in ionic compounds, or shares 2 in molecules like H_2O .

TESTS USUALLY ASK

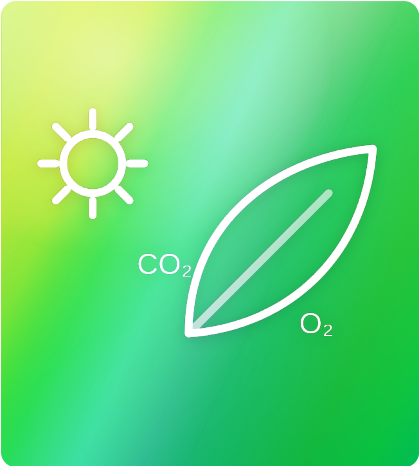
- Electron shells and ions
- Burning = reacting with oxygen



Scan for the full explanation and a practice question.
SPARKON · 07 / 10

Photosynthesis

Fotosyntes



BIOLOGY

Discovered 1779 · Jan Ingenhousz

SPARKON

RARE

10 / 10

The π Mystery

Talet pi



MATH

First estimate 1650 BC · Rhind Papyrus, Egypt

SPARKON

1 OF 1 IN THE WORLD

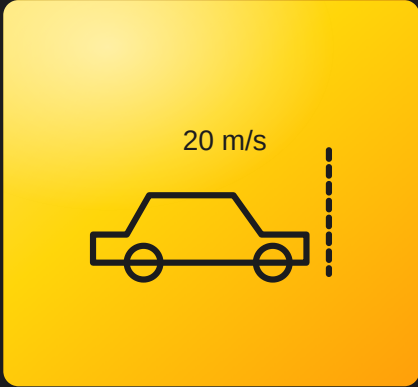
No. 1 / 1

CHALLENGE

LAB DUEL

Stop a car doing 20 m/s within 30 m on a level road.

What braking does it take, and is it possible?



Play a card that solves it and say why in one sentence.

SPARKON

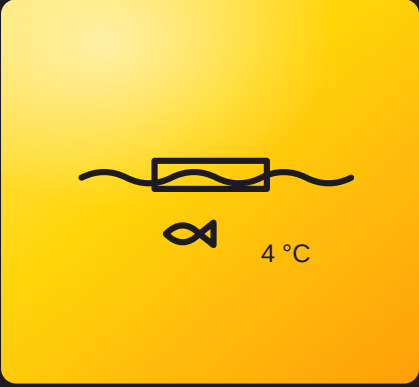
C1

CHALLENGE

LAB DUEL

Why does ice float on a lake?

Explain it so a friend could predict it.



Play a card that solves it and say why in one sentence.

SPARKON

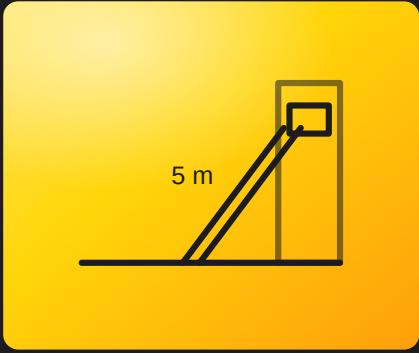
C2

CHALLENGE

LAB DUEL

Will a 5 m ladder reach a window 4 m up if its foot must stand 3 m out?

Yes or no, and prove it.



Play a card that solves it and say why in one sentence.

SPARKON

C3

Accepted answers

C1

Newton's second law

Energy conservation

Two of these together score double.

A WINNING EXPLANATION

Braking must remove the car's movement energy $\frac{1}{2}mv^2$ over 30 m, so it needs about 6.7 m/s^2 of steady braking ($F = m \cdot a$: about 8 kN for a 1,200 kg car). Tyres give that only with good grip, roughly $\mu \geq 0.68$: possible on a good dry road, not on ice. Ignores reaction time.

$$a = v^2 / (2s) = 400 / 60 \approx 6.7 \text{ m/s}^2$$

Close enough counts: the other players decide if the explanation is right.

The π Mystery

π

THE IDEA

Every circle, from a coin to a Ferris wheel, goes round exactly π times its width.

$$\pi = C \div d = 3.14159...$$

LET'S YOU PREDICT

The way round any circle, from one straight measurement.

THE ONLY ONE

Brushed metal, gold-foil π , digits round the edge. Hidden in one random booster pack.



Scan for the full explanation and a practice question.
SPARKON · No. 1 / 1

Photosynthesis

THE IDEA

Plants use light to build sugar from carbon dioxide and water, and give off oxygen.



LET'S YOU PREDICT

Most of a tree's dry mass comes from the air.

TESTS USUALLY ASK

- The balanced formula
- Where the energy comes from



Scan for the full explanation and a practice question.
SPARKON · 10 / 10

Accepted answers

C3

Pythagoras' theorem

Only one card fits this one.

A WINNING EXPLANATION

The ladder is the long side: $3^2 + h^2 = 5^2$, so $h^2 = 16$ and $h = 4 \text{ m}$. It reaches exactly to the window sill, with nothing to spare.

$$\sqrt{(25 - 9)} = \sqrt{16} = 4 \text{ m}$$

Close enough counts: the other players decide if the explanation is right.

Accepted answers

C2

Density

Only one card fits this one.

A WINNING EXPLANATION

Ice (0.92 g/cm^3) is less dense than water (1.00 g/cm^3), so it floats. Going deeper: when water freezes, hydrogen bonds between its molecules lock them into an open pattern with more space.

$$0.92 < 1.00 \rightarrow \text{floats, about } 9/10 \text{ under water}$$

Close enough counts: the other players decide if the explanation is right.

How to make the prototype deck

1. Print pages 1–4 on thick paper (200–300 g/m²) at **actual size / 100 %**, double-sided, **flip on long edge**. Each back is printed behind its front.
2. No duplex printer? Print page 1, put the sheet back in the tray face down, print page 2. Do one test sheet first and hold it against a window to check the fronts and backs line up.
3. Cut along the card edges. The grid is 3 × 3 cards of 63 × 88 mm with no gap, so every cut is a straight line across the sheet. A 3–4 mm corner punch gives the rounded corners of a real trading card.
4. Laminate the cut cards in glossy pouches (or laminate the whole sheet first, then cut, leaving a small edge).
5. Check the QR codes with a phone. They point to <https://sparkon.cards/cards/...>; if the website address changes, rebuild the sheet with the new address.

What is on this sheet

10 concept cards (Math 2 · Physics 3 · Chemistry 3 · Biology 2), the π mystery card and 3 challenge cards for Lab Duel.

Making the rare ones shine

- **Common**: white frame, printed and laminated in normal gloss.
- **♦ Rare**: silver frame and a rainbow sheen over the picture. For real shine, laminate rare cards in **holographic laminating film** instead of clear.
- **★ Legendary**: gold frame. For real gold, print on a laser printer and run **toner-reactive gold foil** through a laminator: the foil sticks only where there is toner.
- **π Mystery**: the paper version here is only the prototype. The real card, one in the world, would be made on a **brushed-aluminium card blank** with π in gold foil and its digits laser-engraved round the edge, then hidden in one random booster pack. (Check prices with a metal business-card maker.)

Lab Duel, quick rules for the test rounds

1. Shuffle the concept cards and deal 3 to each player (2–4 players). Put the challenge cards face down in a pile.
2. Turn over a challenge. Anyone may play a card from their hand and explain in one sentence how it solves the challenge. Say a number if you can.
3. Flip the challenge to check. Right card and a sensible explanation: you keep the challenge. Two accepted cards together score double.
4. Played cards go back into your hand. If nobody solves it, the challenge goes to the bottom of the pile.
5. First to 5 points wins. With only 3 challenges in the prototype, play to 3.

Write down during the test rounds

Which challenge caused the most arguing · which card nobody used · any rule question you had to invent an answer for · how long a round took · whether players looked at the card backs to explain.